
HOMEWORK DAY 24 – *Alternating Series Test and Absolute Convergence §11.5*

1. (a) What is an alternating series?

(b) Under what conditions does an alternating series converge? Answer concisely.

Note: In all problems below you need to make sure to check whether these conditions are satisfied if you are using the alternating series test.

(c) If these conditions are satisfied, what can you say about the remainder $R_n = s - s_n$, where s is the sum of the infinite series and s_n is the n th partial sum?

2. Determine whether the series converges or diverges.

(a) $\frac{2}{3} - \frac{2}{5} + \frac{2}{7} - \frac{2}{9} + \dots$

$$(b) \sum_{k=2}^{\infty} \frac{1}{\ln k}$$

$$(c) \sum_{k=2}^{\infty} (-1)^k \frac{1}{\ln k}$$

$$(d) \sum_{k=1}^{\infty} (-1)^k \frac{3k-1}{2k+1}$$

3. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

(a) $\sum_{n=1}^{\infty} (-1)^n \frac{1}{\sqrt{n+1}}$

(b) §11.5: 22

(c) §11.5: 24

(d) §11.5: 28

(e) §11.5: 34

4. §11.5: 37

5. §11.5: 38

HOMEWORK DAY 25 – *Ratio Test §11.6*

6. §11.6: 1

7. §11.6: 3

8. §11.6: 7

9. §11.6: 8

10. §11.6: 12

11. §11.6: 15

12. Determine whether the following series converge absolutely, converge conditionally, or diverge. State the tests that you used and give a brief explanation.

For example: For problem 11.6:13 you could write “Converges absolutely by direct comparison test, since $|a_n| = |\cos(n\pi/3)/n!| \leq 1/n!$, and $\sum 1/n!$ converges by the ratio test since ... (show it), so $\sum |a_n|$ converges.”

(a)
$$\sum_{k=2}^{\infty} \frac{1}{\sqrt{k+4}}$$

(b)
$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$$

(c) $\sum_{n=3}^{\infty} \frac{4^n}{5^n - 1}$

(d) $\sum_{n=1}^{\infty} (-1)^n \frac{3n-1}{2n+1}$

$$(e) \sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^3 + 1}$$

$$(f) \sum_{n=1}^{\infty} \frac{100^n}{n!}$$

$$(g) \sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$$

HOMEWORK DAY 26 – *Power Series §11.8*

13. Which of the following are power series? Circle the ones that are.

(a) $\sum_{n=0}^{\infty} (3x)^n$ (b) $\sum_{n=0}^{\infty} \sqrt{x^n}$ (c) $\sum_{n=0}^{\infty} (x+2)^{2n}$

14. Find the radius of convergence and the interval of convergence of the power series

(a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

(b) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$(c) \sum_{n=10}^{\infty} \frac{(x+1)^n}{n^3 3^n}$$

$$(d) \sum_{n=0}^{\infty} \frac{(x-3)^n}{n^n}$$

$$(e) \sum_{n=1}^{\infty} (-1)^n \frac{2^n (x-2)^n}{n^{1/3}}$$

$$(f) \sum_{n=100}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

15. Suppose $\sum_{n=1}^{\infty} a_n 3^n$ converges and $\sum_{n=1}^{\infty} a_n 6^n$ diverges. What can you say about

(a) the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$?

(b) the series $\sum_{n=1}^{\infty} a_n 2^n$?

(c) the series $\sum_{n=1}^{\infty} a_n$?

(d) the series $\sum_{n=1}^{\infty} a_n 8^n$?

(e) the series $\sum_{n=1}^{\infty} a_n (-3)^n$?

(f) the interval of convergence of the power series $\sum_{n=1}^{\infty} a_n (x - a)^n$?